

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2– Case Study (2 QUESTIONS)

Course Code:	ITA05	Course Title:	COMPUTER VISION
CO Assessed:	CO3 – Precision (BL4)	Assessment:	Assessment Tool 1 – Case Study
Weightage:	40% of CIA	Total Marks:	20 (2 Questions × 10Marks)
CO1 Statement:	<i>Analyze and extract image features using detection and matching techniques for effective image representation and comparison..(BL4)</i>		
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Section / Batch:	2024-slot D	Date:	10.8.2024

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Code of Conduct

I T.Chaitra(192425217) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:T.Chaitra

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Feature Detection and Matching	Preprocessing, Edge Detection, Harris Corner Detection, Hough Transform, SIFT	1	10
Feature Detection and Matching	PCA, Image Representation, Feature Matching, Image Processing Applications	2	10
GRAND TOTAL:		20 Marks	

Annexure A –Case Study

1. A remote sensing application is used to align satellite images of the same geographical region captured at different times for monitoring environmental changes. The system initially relies on edge detection to identify matching regions between the images; however, due to differences in illumination, image orientation, and background details, the matching process becomes unreliable and produces inaccurate alignment results. The development team decides to replace the edge-based approach with Harris Corner Detection to identify stable and distinctive feature points. Analyze the limitations of edge detection in this application and explain how Harris Corner Detection improves feature localization, matching accuracy, and robustness for image registration under varying imaging conditions.

2. An autonomous surveillance system continuously captures high-resolution color images to detect vehicles and pedestrians in real time. Processing the original color images directly increases computational complexity and slows down feature extraction, making it difficult to achieve the required processing speed. To improve system performance, the engineers consider converting the images into grayscale or binary representations before applying feature detection algorithms. Analyze the impact of different image representations on computational efficiency, edge detection, and feature extraction performance, and justify the most appropriate image representation for achieving accurate and reliable object detection in real-time computer vision applications.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Faculty In-charge
Signature: _____
Date: _____

1. Harris Corner Detection for Satellite Image Registration

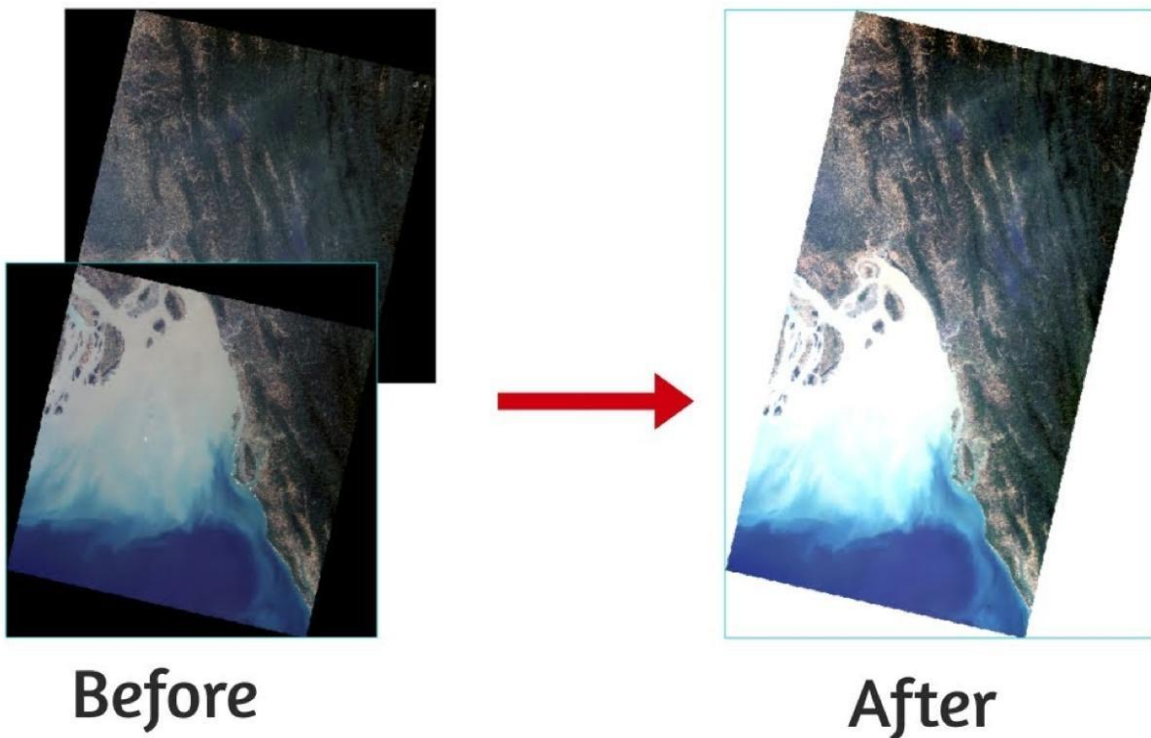
1. Introduction

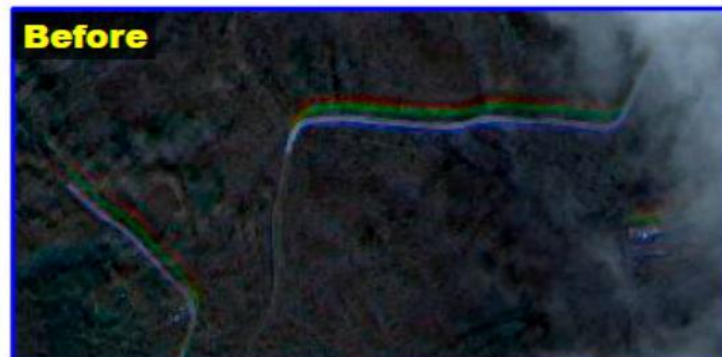
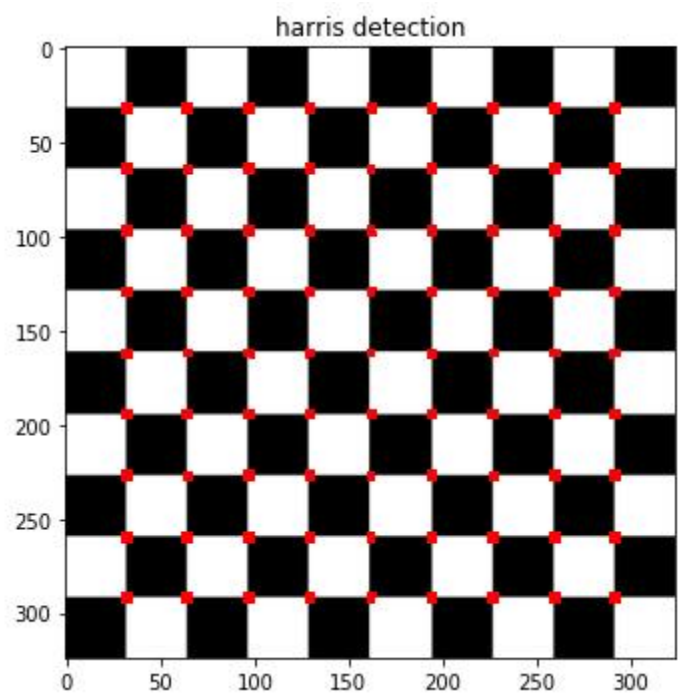
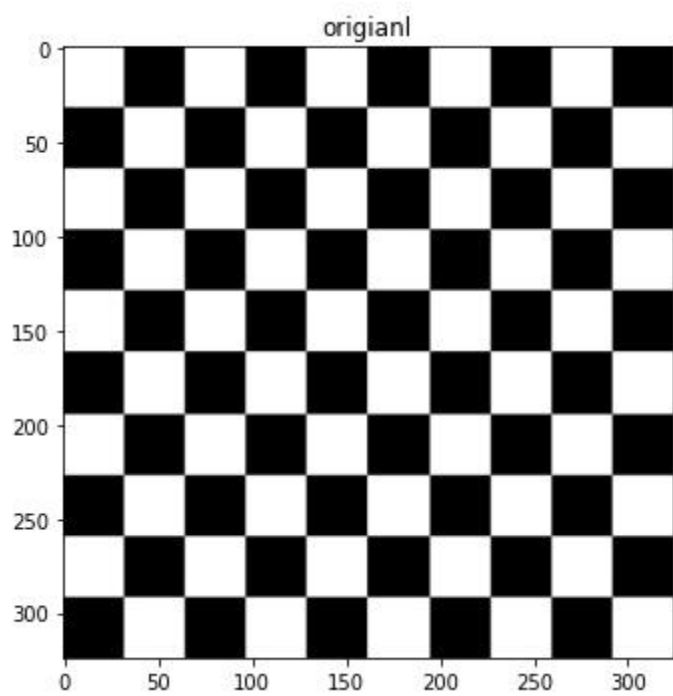
Remote sensing is an important application of computer vision in which satellite images are used to observe and analyze the Earth's surface. Satellite images of the same geographical region may be captured at different times to monitor environmental changes such as deforestation, urban development, floods, crop growth, land-use changes, and changes in geographical boundaries.

Before comparing two satellite images, the images must be accurately aligned with each other. This process is called image registration. Image registration means transforming one image so that the same geographical locations in both images overlap correctly.

Initially, the given system uses edge detection to identify matching regions between satellite images. However, edge detection alone may not provide reliable matching because edges can be affected by illumination changes, image orientation, noise, shadows, and background variations.

Therefore, the development team replaces the edge-based method with Harris Corner Detection. Harris Corner Detection identifies distinctive points called corners, which provide better feature localization and more reliable matching between images.





2. Image Registration in Remote Sensing

Image registration is the process of aligning two or more images of the same scene captured at different times, viewpoints, or imaging conditions.

For example:

Satellite Image 1 → January

Satellite Image 2 → June

Both images may contain the same:

- Buildings
- Roads
- Rivers
- Agricultural fields
- Mountains
- Forest boundaries

But their appearance may be different because of changes in lighting, weather, camera orientation, and background details.

Basic Image Registration Process

Input



Feature



Feature



Transformation



Image Alignment / Registration

The main requirement is to find stable and distinctive feature points that appear in both images.

3. Edge Detection

Edge detection identifies locations where there is a sudden change in image intensity.

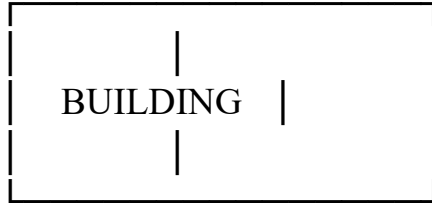
For example, edges can occur at:

- Building boundaries
- Road boundaries
- Forest boundaries
- River boundaries
- Mountain boundaries
- Object boundaries

An edge detector can therefore provide useful structural information from a satellite image.

Example

A building may appear as:



The boundaries of the building produce strong edges.

However, an edge is generally a line or continuous boundary, rather than a single distinctive point.

This creates problems when two images need to be matched accurately.

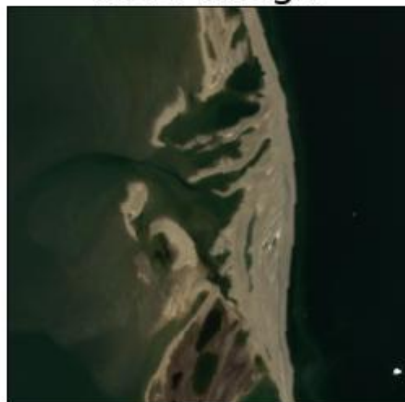


Original

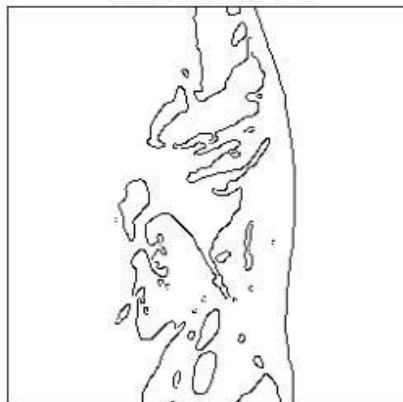


Data Range (3 x 3 kernel)

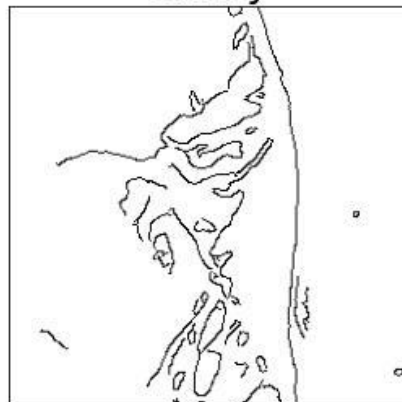
RGB Image



Reference



Canny



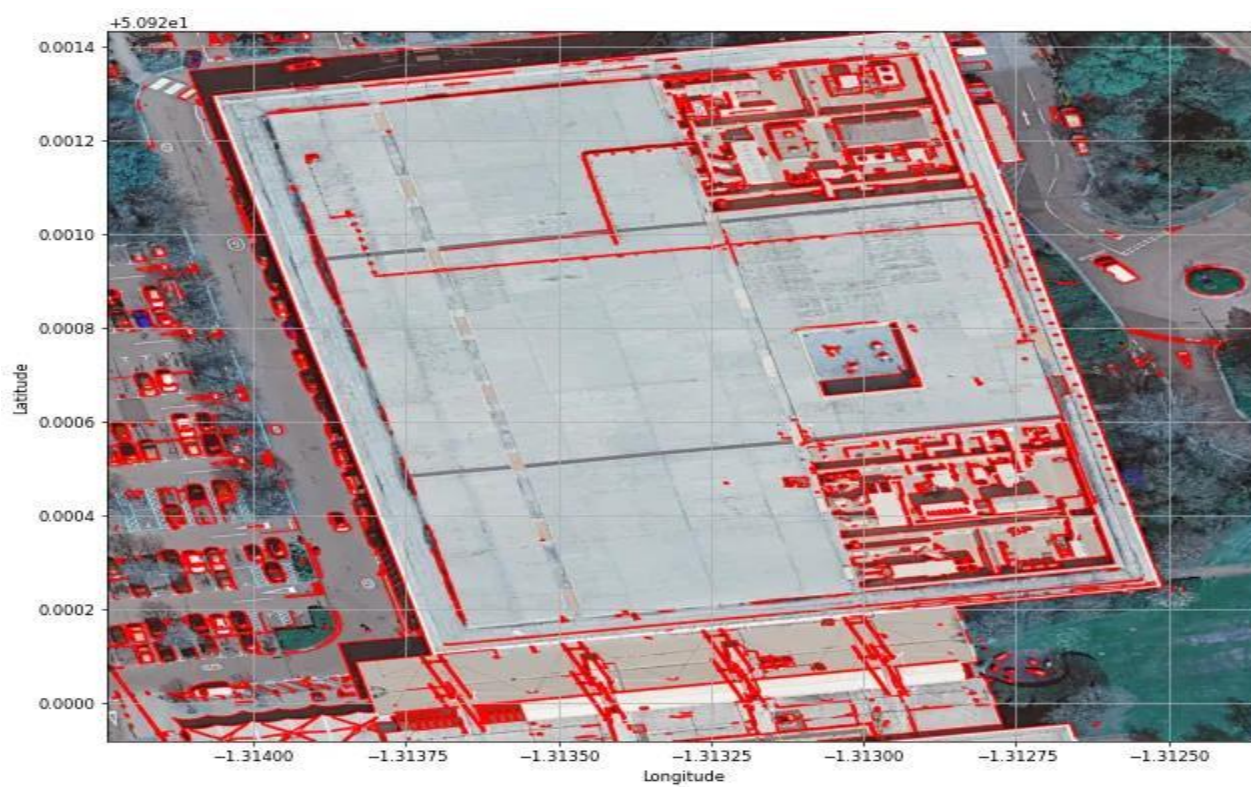
Sobel



Scharr



Prewitt



4. Limitations of Edge Detection

Although edge detection is useful for finding boundaries, it is not ideal as the primary feature for satellite image registration.

Sensitivity to Illumination

Satellite images captured at different times may have different illumination conditions.

For example:

- Morning → low sunlight
- Afternoon → strong sunlight
- Cloudy weather → reduced brightness
- Different seasons → different shadows

These changes can modify the appearance of edges.

A building may produce a strong edge in one image but a weaker edge in another image.

Therefore, direct edge matching may produce incorrect correspondences.

5. Harris Corner Detection

Harris Corner Detection is a feature detection technique used to identify **corners or interest points** in an image.

A corner is a location where the image intensity changes significantly in **two different directions**.

For example, the corner of a building contains changes in both horizontal and vertical directions.



The point where the horizontal and vertical boundaries meet is a corner.

Harris Corner Detection focuses on these distinctive points rather than using complete edge lines.

6. Working Principle of Harris Corner Detection

The basic process can be represented as:

Input Image



Grayscale Conversion



Calculate Image Gradients



Structure Matrix



Harris Response Calculation



Select Strong Corners

Step 1: Grayscale Conversion

The input satellite image is converted into grayscale so that intensity changes can be analyzed.

Step 2: Calculate Gradients

The system calculates intensity changes in horizontal and vertical directions.

These gradients help identify areas where image intensity changes strongly.

Step 3: Structure Matrix

The gradient information is represented using a structure matrix.

Step 4: Harris Response

A Harris response value is calculated for each location.

A large positive response indicates a strong corner.

Step 5: Select Strong Corners

The points with strong Harris responses are selected as important feature points

7. Robustness to Illumination Changes

Satellite images captured at different times can have different brightness levels.

For example:

Image A

Image B

☀ Bright

☁ Less bright

Building



Building



Even though the brightness changes, important structural features such as:

- Building corners
- Road intersections
- Field boundaries
- Structural junctions

may remain useful.

Harris Corner Detection uses **local intensity gradients** rather than simply comparing absolute pixel brightness.

Therefore, it provides better robustness when illumination conditions change.

8. Robustness to Background Details

Satellite images contain many background objects that may produce unwanted edges.

For example:

Trees + Clouds + Shadows + Water + Buildings

may all generate edges.

Edge detection can therefore produce many irrelevant features.

Harris Corner Detection instead focuses on points where intensity changes occur in multiple directions.

This reduces ambiguous features and makes the matching process more reliable.

9. Handling Geometric Transformations

Satellite images may experience different geometric transformations such as:

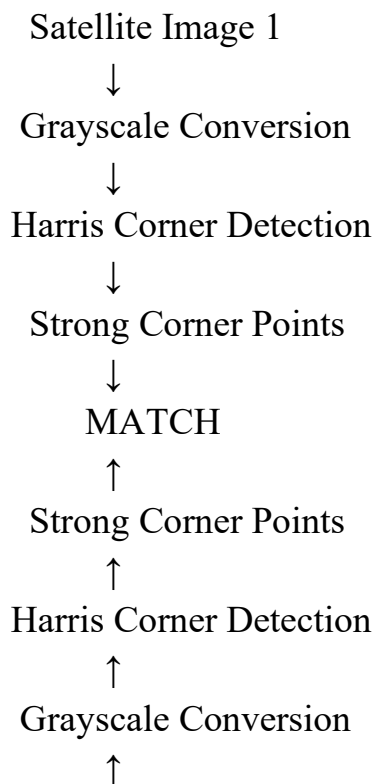
- Translation
- Rotation
- Scaling
- Slight perspective changes

Harris corner points provide strong reference locations.

These points can be used to estimate the transformation required to align the images.

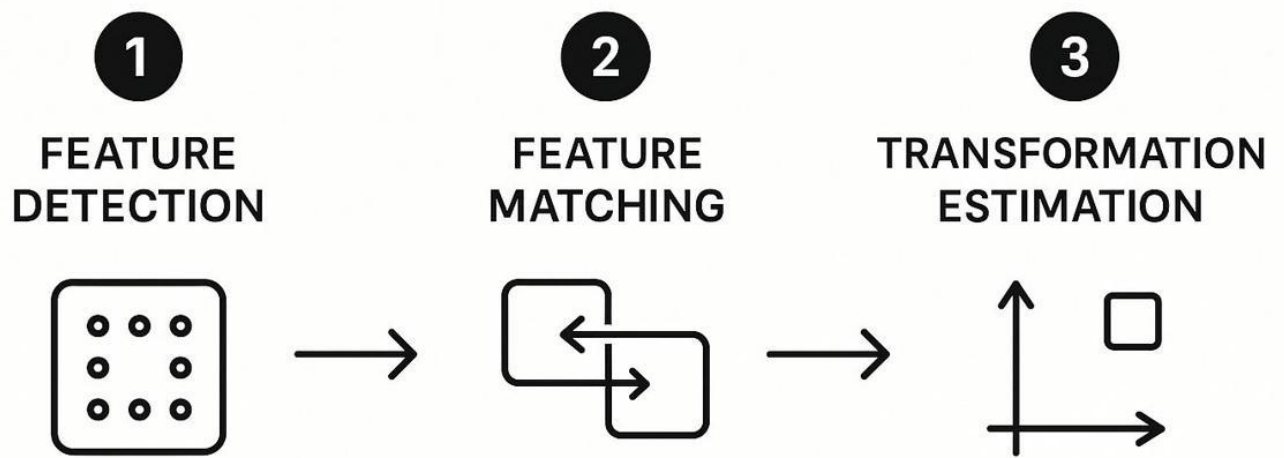
10. Overall Image Registration Pipeline

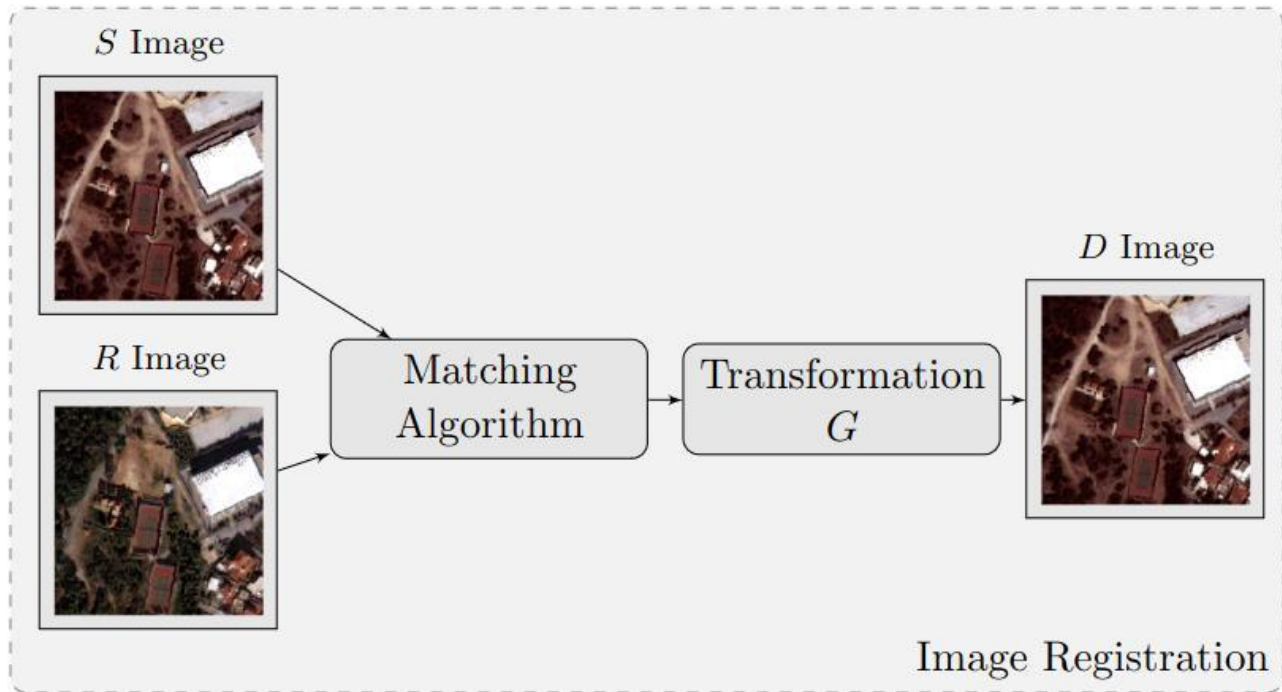
The complete improved system can be represented as:



Satellite Image 2
↓
Transformation Estimation
↓
Image Registration
↓
Environmental Analysis

IMAGE REGISTRATION IN PYTHON





Conclusion

Edge detection is useful for identifying boundaries in satellite images, but it is not ideal as the main feature for image registration. Edges are often long, repetitive, sensitive to illumination, affected by background variations, and difficult to localize precisely.

Harris Corner Detection solves many of these problems by identifying stable and distinctive points where image intensity changes in two directions.

These corner points provide:

- Better feature localization
- More accurate feature matching
- Improved robustness to illumination changes
- Reduced ambiguity
- Better support for geometric transformations
- More reliable image registration

Therefore, replacing edge detection with Harris Corner Detection significantly improves the accuracy and reliability of satellite image registration. This makes it more suitable for applications such as deforestation monitoring, flood detection, urban growth analysis, crop monitoring, and land-use change detection.

2. Choosing Grayscale or Binary Images for Real-Time Object Detection

1. Introduction

An autonomous surveillance system continuously captures **high-resolution color images** to detect objects such as **vehicles and pedestrians** in real time.

In a surveillance system, the captured images have to be processed very quickly. If the original RGB color images are directly given to feature detection algorithms, the system has to process information from **three color channels** for every pixel. This increases:

- Computational complexity
- Memory usage
- Processing time
- Feature extraction time

Therefore, the engineers consider converting the original color images into either **grayscale** or **binary** representations.

The choice of image representation has a direct effect on:

1. Computational efficiency
2. Edge detection
3. Feature extraction
4. Object details
5. Detection accuracy
6. Real-time processing speed

For general real-time vehicle and pedestrian detection, **grayscale representation is the most appropriate choice** because it provides a good balance between computational efficiency and detection accuracy.

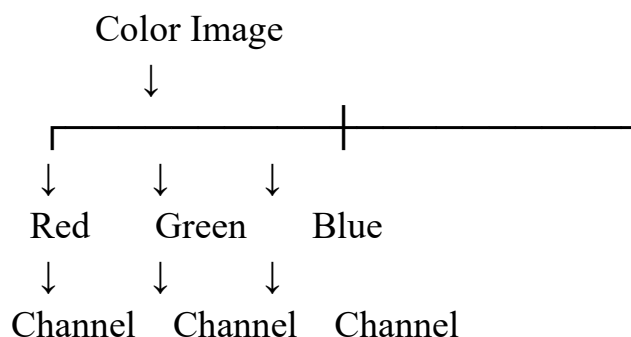
2. Color Image Representation

A normal color image is represented using three channels:

- Red (R)
- Green (G)
- Blue (B)

Therefore, every pixel contains information from three channels.

For example:



For a high-resolution image, processing all three channels requires more computational

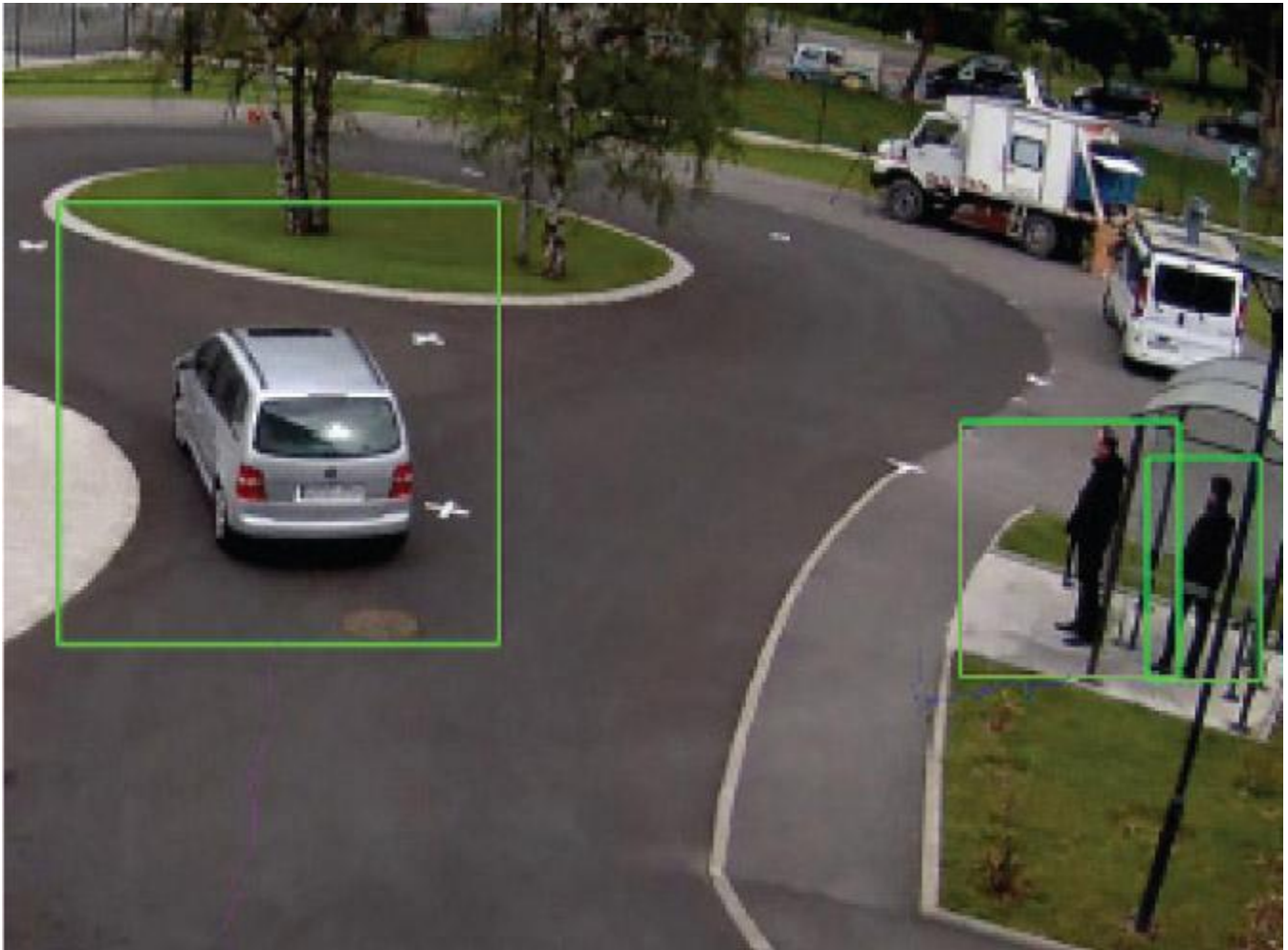
resources.

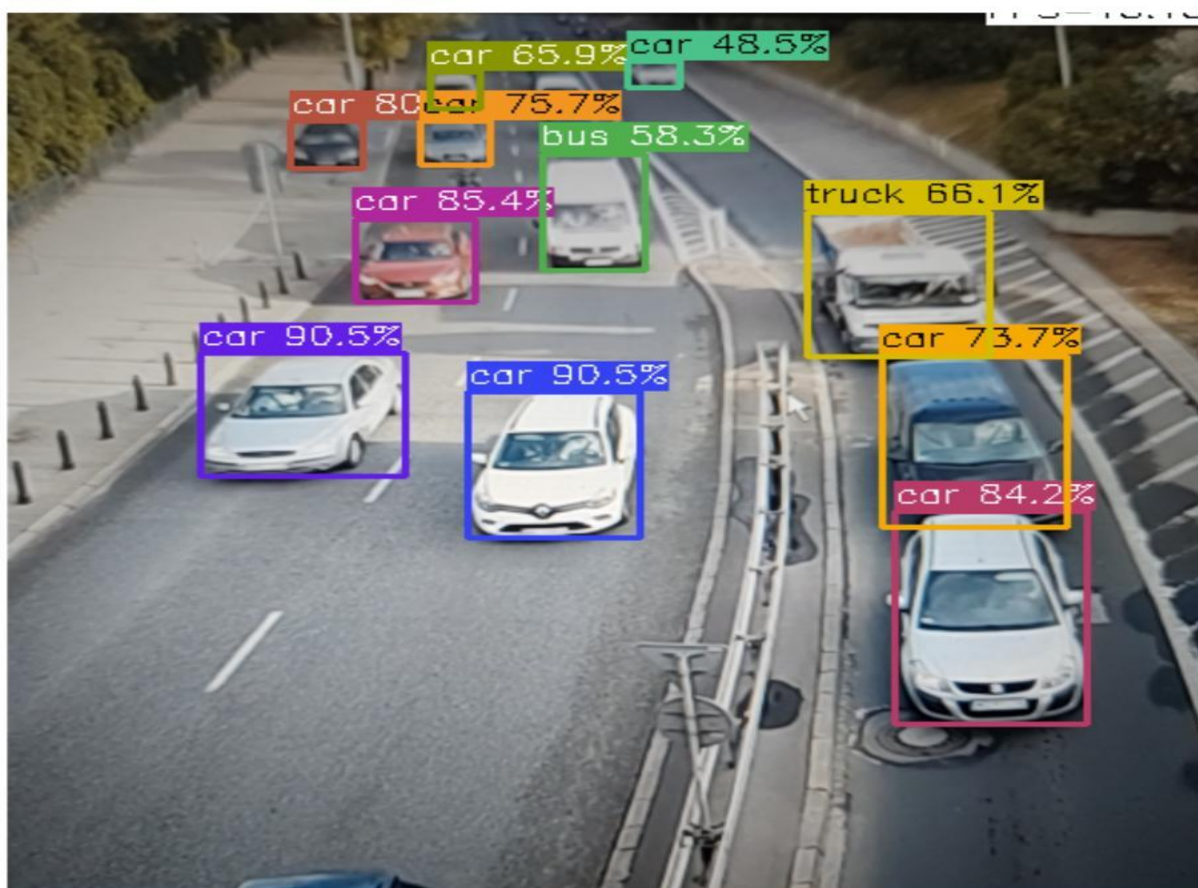
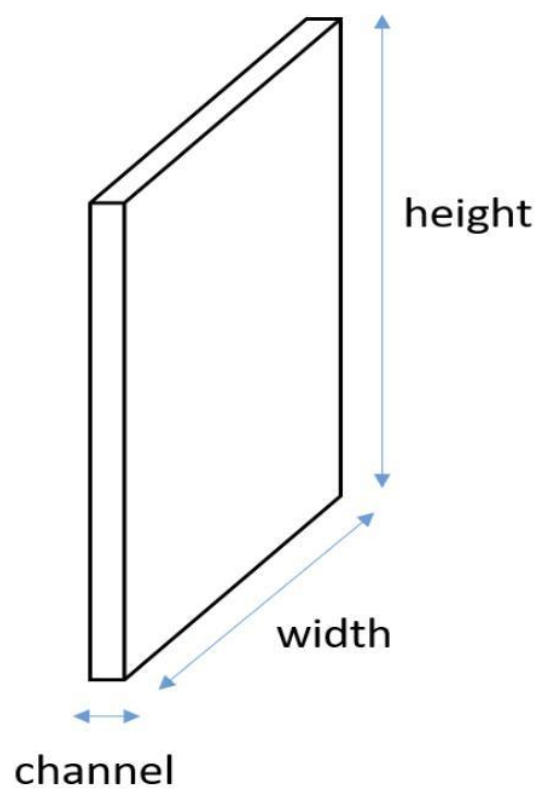
Main disadvantages

- Higher memory requirement
- Higher processing time
- More data to process
- Increased feature extraction complexity

However, color information can sometimes help distinguish objects from their surroundings.

For example, a **red vehicle against a green background** can be easily separated using color information.





3. Grayscale Image Representation

A grayscale image contains only **one intensity channel** instead of three color channels.
The intensity normally ranges from:

0 → Black

255 → White

Color Image



Grayscale Conversion



Single Intensity Channel



Feature Detection

Instead of processing R, G, and B separately, the system processes only one intensity value for each pixel.

Therefore, grayscale representation significantly reduces the amount of data that needs to be processed.

4. Advantages of Grayscale Representation

4.1 Reduced Computational Complexity

RGB images contain three channels, whereas grayscale contains only one channel.

Therefore:

RGB

$R + G + B$



More data



Higher computation

Grayscale

Intensity



Less data



Lower computation

Only one channel needs to be processed.

This makes grayscale suitable for real-time applications.

4.2 Faster Feature Extraction

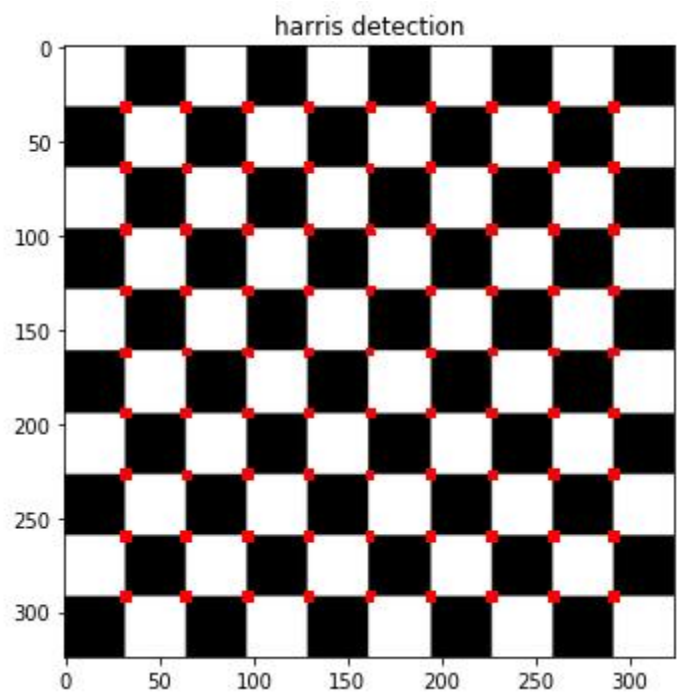
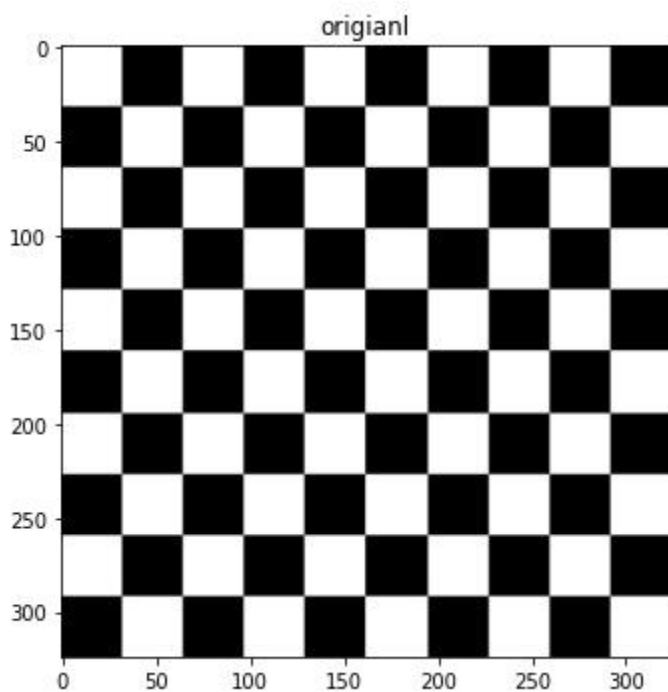
Many computer vision feature detection algorithms can operate effectively on grayscale images.

Examples include:

- Harris Corner Detection
- SIFT
- Canny
- Sobel
- HOG

Since only one channel is processed, feature extraction becomes faster.

The case study specifically identifies these algorithms as suitable for grayscale images.

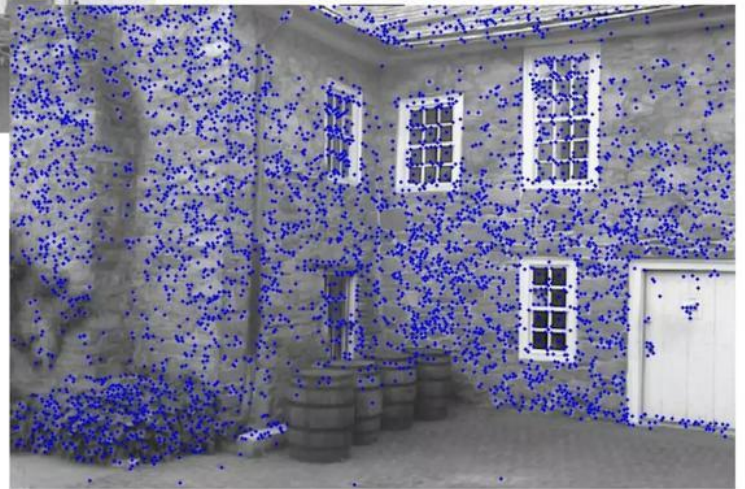


SIFT Examples



← Original

Applying SIFT



5. Grayscale and Edge Detection

Edge detection mainly depends on **changes in image intensity**.

For example:

Dark Region



← Edge



Bright Region

Because grayscale directly represents intensity, it works very well for edge detection.

A common example is the **Canny Edge Detector**, which normally operates on a grayscale image.

Other techniques such as **Sobel** can also detect intensity changes efficiently.

Therefore, grayscale provides good edge detection performance while requiring less

computational power than RGB processing.

6. Memory Requirement of Grayscale

Grayscale images require less memory than RGB images.

For an RGB image:

3 channels $\rightarrow$ R + G + B

For grayscale:

1 channel $\rightarrow$ Intensity

Therefore, grayscale requires approximately **one-third of the channel data** compared with an RGB image.

This is especially useful for high-resolution surveillance cameras where thousands of images may need to be processed continuously.

Conclusion

RGB images preserve maximum image information, but processing three color channels increases computational complexity and memory usage.

Binary images are extremely fast to process, but they lose significant image information and can be affected by threshold selection and changing illumination.

Grayscale representation is the most appropriate choice for the given real-time surveillance system. It reduces computational requirements while retaining sufficient intensity information for reliable edge detection and feature extraction.

Therefore, the recommended pipeline is:

RGB Image $\rightarrow$ Grayscale $\rightarrow$ Noise Reduction $\rightarrow$ Edge/Feature Detection $\rightarrow$ Object Detection $\rightarrow$ Vehicle/Pedestrian Recognition

Thus, grayscale provides the best overall compromise between computational efficiency, feature extraction performance, and accurate and reliable real-time object detection.